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CONTRIBUTION

Source:	Ministry of Communications, India; Centre for Development of Telematics, India
Title:	Proposed baseline text for the proposed draft new ITU-T Technical Report JSTR.WIFI-TV "Secondary distribution of digital television and audiovisual content to portable devices using WiFi"

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Abstract: This contribution contains the draft baseline text for the proposed draft new ITU-T Technical Report JSTR.WIFI-TV on Secondary distribution of digital television and audiovisual content to portable devices using WiFi and other Wireless Local Area Networks.

Summary and Rationale: The proposed Technical Report will address aspects related to delivering television and other audiovisual signals to smartphones and other portable devices without using the mobile data and the mobile networks, while using WiFi or other Wireless Local Area Networks (WLANs) for secondary distribution and the conventional modes of satellite, cable, or terrestrial transmission for primary distribution. The proposed Technical Report would cover the network architecture, transcoding requirements, other technical requirements, mechanism to facilitate local content distribution, local advertisement insertion, logging end user statistics, case studies, and more.

The objective of the proposed Technical Report is to enhance the use of widely available WiFi networks and other Wireless LAN networks for the last mile delivery of linear television and audiovisual content to portable devices in a manner that is cost-effective for the end users and does not load the mobile networks.

Proposal: The draft baseline text for the proposed draft new ITU-T Technical Report JSTR.WIFI-TV is placed as ANNEX-1.

I n t e r n a t i o n a l   T e l e c o m m u n i c a t i o n   U n i o n

ITU-T Technical Report

TELECOMMUNICATION
STANDARDIZATION SECTOR
OF ITU

(06 SEP 2022)

JSTR.WIFI-TV
Secondary distribution of digital television and audiovisual content to portable devices using WiFi

ITU-T

International
Telecommunication
Union

Summary

This technical report addresses aspects related to delivering television and other audiovisual signals to smartphones and other portable devices without using the mobile data and the mobile networks, while using WiFi or other Wireless Local Area Networks (WLANs) for secondary distribution and the conventional modes of satellite, cable, or terrestrial transmission for primary distribution.

This concept has high value propositions in rural areas, public utility places such as railway stations and airports, and even in moving vehicles such as buses, taxis, etc.

The report covers the network architecture, transcoding requirements, other technical requirements, mechanism to facilitate local content distribution, local advertisement insertion, logging end user statistics, and more. The report also provides global best practices through case studies.

Keywords

Converged assess node, WiFi, secondary distribution, smartphone

Change Log

This document contains Version 1 of the ITU-T Technical Report JSTR.WIFI-TV on "Secondary distribution of digital television and audiovisual content to portable devices using WiFi" approved at the ITU-T Study Group 09 meeting held in Geneva, 06-14 Sept. 2022.

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ITU-T Technical Report on Secondary distribution of digital television and audiovisual content to portable devices using WiFi

Summary

The Technical Report addresses aspects related to delivering television and other audiovisual signals to smartphones and other portable devices without using the mobile data and the mobile networks, while using WiFi or other Wireless Local Area Networks (WLANs) for secondary distribution and the conventional modes of satellite, cable, or terrestrial transmission for primary distribution.

This concept has high value propositions in rural areas, public utility places such as railway stations and airports, and even in moving vehicles such as buses, taxis, etc.

The Report covers the network architecture, transcoding requirements, other technical requirements, mechanism to facilitate local content distribution, local advertisement insertion, logging end user statistics, and more. The Report also provides global best practices through case studies.

1 Scope

The Technical Report addresses aspects related to delivering television and other audiovisual signals to smartphones and other portable devices without using the mobile data and the mobile networks, while using WiFi or other Wireless Local Area Networks (WLANs) for secondary distribution and the conventional modes of satellite, cable, or terrestrial transmission for primary distribution.

The objective of the Technical Report is to enhance the use of widely available WiFi networks and other Wireless LAN networks for the last mile delivery of linear television and audiovisual content to portable devices in a manner that is cost-effective for the end users and does not load the mobile networks.

2 References

[1] To be added as the study progresses.

3 Terms and definitions

3.1 Terms defined elsewhere

This Technical Report uses the following terms defined elsewhere:

3.1.1 term [reference]: definition

3.1.2 term [reference]: definition

3.2 Terms defined here

This Technical Report defines the following terms:

3.1.1 term [reference]: definition

3.1.2 term [reference]: definition

[To be added as the study progresses.]

4 Abbreviations

WLAN Wireless Local Area Network

5 Introduction

Mobile handsets, tablets and laptops are becoming the preferred devices for viewing linear television and audiovisual content. This however consumes a lot of mobile data and is costly for the consumers. Also, many concurrent users may create network congestion in typical mobile networks, thus limiting the Quality of Service. Harnessing the digital satellite and digital cable platforms requires the receiving device to have an integrated or separate DVB-T2/S2 receiver module. This is not feasible for smartphones and other mobile devices.

The solution exists in using a converged access node as a middleware, which receives the linear television and other signals through the conventional modes such as satellite, cable or terrestrial, and then distributes them using WiFi or other WLAN. The end users can view the television or other audiovisual content on their portable devices without consuming mobile data and without requiring any additional hardware or plugin, etc.

This concept has high value propositions in rural areas, public utility places such as railway stations and airports, and even in moving vehicles such as buses, taxis, etc.

The proposed Technical Report aims to provide guidance on technical aspects and also share global best practices through case studies.

6 Network architecture for secondary distribution using WiFi or WLAN

[Editor's note: Text to be drafted – contributions encouraged.]

7 Converged access node requirements

[Editor's note: Text to be drafted – contributions encouraged.]

8 Transcoding requirements

[Editor's note: Text to be drafted – contributions encouraged.]

9 Requirements to meet regulatory compliance for encrypted channels

[Editor's note: Text to be drafted – contributions encouraged.]

10 Local content distribution

[Editor's note: Text to be drafted – contributions encouraged.]

11 Local advertisement insertion

[Editor's note: Text to be drafted – contributions encouraged.]

12 End user statistics logging

[Editor's note: Text to be drafted – contributions encouraged.]

13 Case studies and global best practices

[Editor's note: Text to be drafted – contributions encouraged.]

14 Concluding Remarks

[Editor's note: Text to be drafted – contributions encouraged.]

15 Bibliography

[To be added as the study progresses.]